



ZEUS ENGINEERING JSC

SSMDC

Small Solar Modular Datacenter

Decentralized AI compute.
Deployed in months, not years.



Hyperscale cannot keep up with the hunger for compute.

Years of Delay

Paperwork, permits, grid interconnection and commissioning routinely stretch 3–7 years before a hyperscale campus delivers a single FLOP.

Concentrated Risk

Single-site failure — power, cooling, network or regulation — takes down massive capacity. Uptime is tied to one location.

Capital & Site Lock-in

Hundreds of millions locked into one geography. Land, power contracts and local politics become single points of failure.

Where the modular approach wins

Capability	Hyperscale	SSMDC
Deployment speed	✗ 3–7+ years	✓ Months
Capital intensity (first capacity)	✗ Hundreds of millions	✓ Sub-\$1M per node
Site / single-point risk	✗ High	✓ Low (distributed nodes)
Uptime model	✗ Tied to one location	✓ Network of nodes
On-site solar + battery	✗ Rare / limited	✓ Core design
DC-DC power efficiency	✗ Usually AC-heavy	✓ 8–15% ongoing savings
Real-estate offset model	✗ Pure compute asset	✓ Land + compute dual asset
Time to first revenue	✗ Years	✓ Far shorter
Scalability approach	✗ Big-bang / multi-year	✓ Add nodes as needed
Existing live prototype	✗ —	✓ 100 kWp operational

SSMDC — Small Solar Modular Datacenter

A decentralized, modular, cost-effective alternative to hyperscale.
Address site risk, slash deployment time, and ensure network uptime by nodes — not by site.

01 Modular

Factory-built or rapid-deploy units. Scale by adding nodes.

02 Solar + Battery

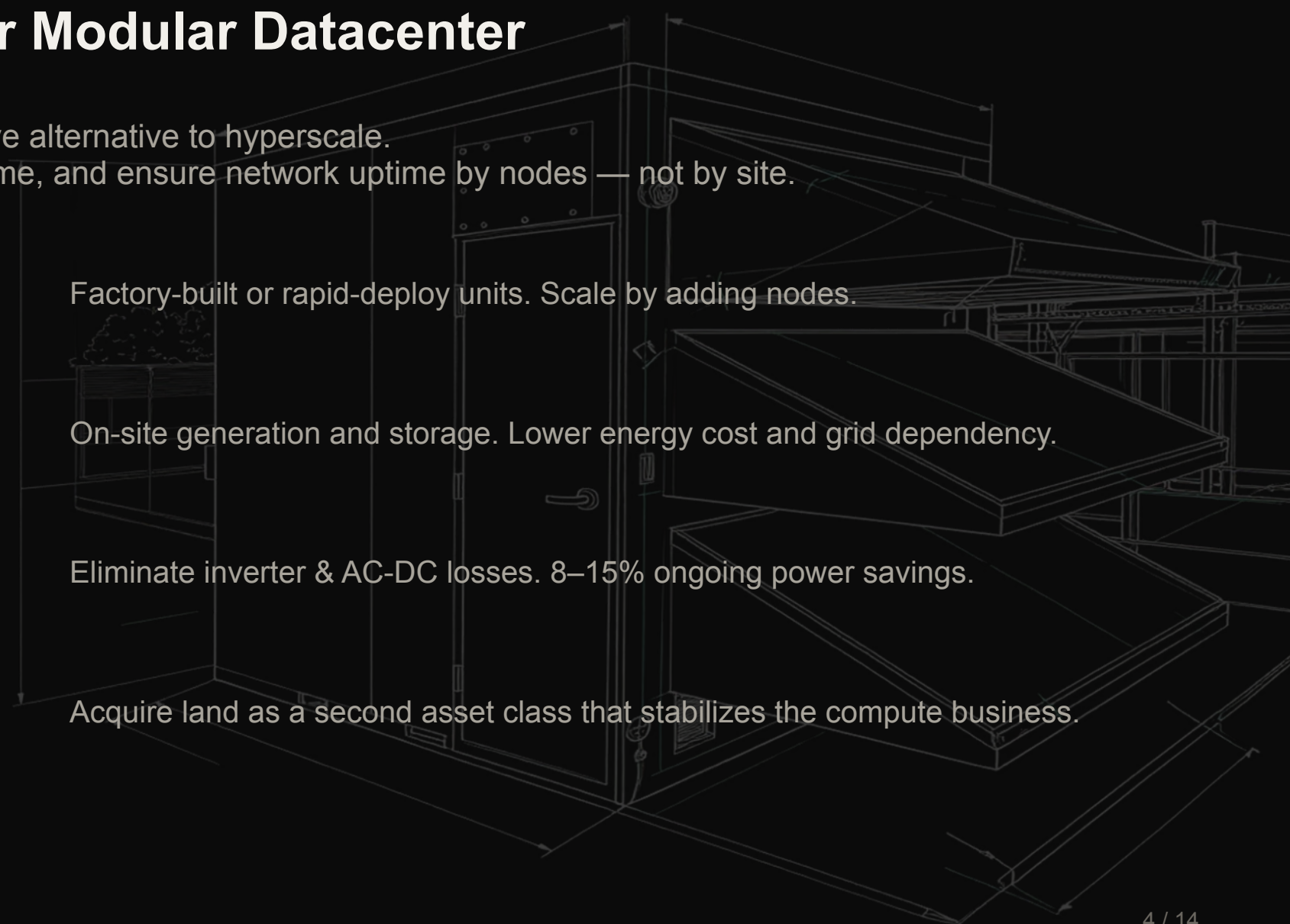
On-site generation and storage. Lower energy cost and grid dependency.

03 DC-DC Architecture

Eliminate inverter & AC-DC losses. 8–15% ongoing power savings.

04 Real-Estate Offset

Acquire land as a second asset class that stabilizes the compute business.



Full-stack efficiency from sun to silicon

DC-DC POWER PATH

Solar → Battery → High-efficiency DC distribution → DC-input compute

Removes two major conversion stages (inverter + server AC-DC PSUs). Real-world facility-level savings of 8–15%, with a practical 10–12% central case. Less heat, less copper, simpler protection at modular scale.

MODULAR + SOLAR + STORAGE

400 m² class sites become viable AI nodes

80–120 kWp solar, 300–500 kWh battery, 50–75 kW continuous IT load in strong solar regions such as Bà Rịa. Deployable far faster than campus builds. Network resilience comes from many independent nodes rather than one large site.

Already operating. Already learning.

5 yrs

Established in Vietnam

100 kWp

Operational ASIC prototype

BTC

Live compute workload

DC path

Architecture validated

Zeus Engineering has spent a decade building dense power, cooling and modular compute systems — first for mining, now redirected to AI. The Vietnam prototype is the foundation for the next generation of SSMDC nodes.

400 m² class node — Bà Rịa region, South Vietnam

Solar capacity	80–120 kWp (realistic working range)
Annual solar yield	~130–190 MWh (strong southern resource)
Battery	300–500 kWh usable (3–5 h class)
Continuous IT load	50–75 kW average, higher peaks with storage
Power architecture	End-to-end DC-DC — 8–15% ongoing savings
Deployment model	Modular / container + outdoor energy plant

Recommended first-node configuration

24–32

NVIDIA H100 / H200 GPUs

~50–100+

PFLOPS (FP8 sparse)

\$0.9–1.4M

GPU hardware cost

Fits cleanly inside the 50–75 kW continuous IT envelope. Delivers meaningful production inference and medium-scale training / fine-tuning capacity.

Memory

1.9–4.5 TB HBM

Inference (70B-class)

30–100k+ tokens/s

Form factor

3–4 × 8-GPU servers

Upgrade path

B200 when ready

H200 preferred for memory-heavy workloads. Configuration stays modular and can be expanded by adding nodes rather than enlarging a single site.

Two asset classes. One operating system.

COMPUTE

AI training & inference capacity sold as nodes or managed service. High utilization, recurring revenue. Energy cost structurally lower via solar + DC-DC path.

REAL ESTATE

Land and buildings acquired as a core asset. Appreciating (or cash-flowing) property offsets engineering and operations cost — the McDonald's model applied to modular compute.

Full-stack operators, not pure software or pure real-estate.

Power systems DNA

Electrical & electronics engineering (UNSW Canberra) + decade of dense GPU/ASIC cluster operations for mining.

Military systems discipline

EE (UNSW Canberra — #1 Australia / #19 global QS). 20+ years experience. Exceeds Vietnam Govt high-tech expert criteria. Blackhawk & MRH-90 systems background — weight, power, thermal & reliability under constraint.

Live prototype

100 kWp BTC ASIC site already running in Vietnam. Real data, real lessons.

Vertical integration

From solar + battery design through DC distribution to compute packaging and site acquisition.

An ecosystem for distributed AI infrastructure



OpenAI



Lazada



EVN



Shopee

From proven prototype to distributed network

NOW

Vietnam Prototype

100 kWp ASIC site operational
DC-DC and solar+battery lessons captured
Team and process in place

NEXT

SSMDC v1 Nodes

Standardized 400 m² class modules
Bà Rịa and additional southern sites
First AI workloads alongside mining

SCALE

Network Expansion

Australia company vehicle
Multi-node portfolio
Real-estate + compute balance sheet

THE ASK

\$2.69M · 18–24 month runway to multiple live nodes

Use of Funds	Amount	Purpose
GPU & Server Hardware	\$1.15M	2 nodes: 24–32× H200-class GPUs + servers, NVLink, networking, DC-input PSUs
Founders, Staff & Contractors	\$620k	Fair living wages for 2 founders + engineers, installers, ops, legal (18–24 mo, taxes & SI included)
Solar, Battery, Power & Build-out	\$380k	80–120 kWp solar + 300–500 kWh battery + DC plant + containers/site works for 2–3 nodes
Land Options, Deposits & Setup	\$180k	Site options in Bà Rịa region + permits, corporate, import clearance & legal
R&D, Taxes, Contingency & Working Capital	\$360k	DC-DC R&D, import duties/VAT buffer, insurance, legal, runway contingency

Founder living wages set at competitive senior technical rates in Vietnam (net ~\$5.5–8k / \$4.5–6.5k monthly) with full employer taxes & social insurance included. High-tech / AI expert incentives may apply.



ZEUS ENGINEERING JSC

Build the nodes. Own the uptime.

SSMDC — Small Solar Modular Datacenter
Decentralized. Modular. Deployable.

Contact · Zeus Engineering JSC · Vietnam

